



Linear equations in one unknown: Reconstruction

An equation constrains an unknown

Subject: Mathematics

Topic: Linear equations

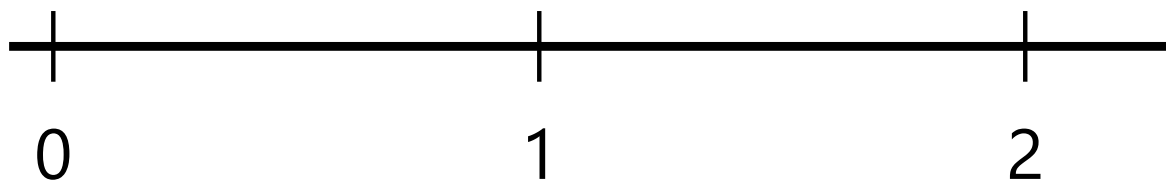
Concept: An equation constrains an unknown

Concept difficulty: Medium concept

Predict

Before you change anything in $3x + 2 = 9$: does this line ask you to do something, or does it claim something? Commit to one, and say what the other reading would mean.

Domain: the rationals · x (dimension



Declared domain

Attempt

Choose one value of x that you believe satisfies $3x + 2 = 9$ and one that you believe does not. For each, substitute into the equation and write both sides separately. Then say which, if either, is a solution and why.

- Use rational candidates to test whether $3x + 2 = 9$ is true.

Boundary test

Is $0x + 2 = 9$ a statement about an unknown? Which values of x make it true?

Operations that preserve the solution set

Subject: Mathematics

Topic: Linear equations

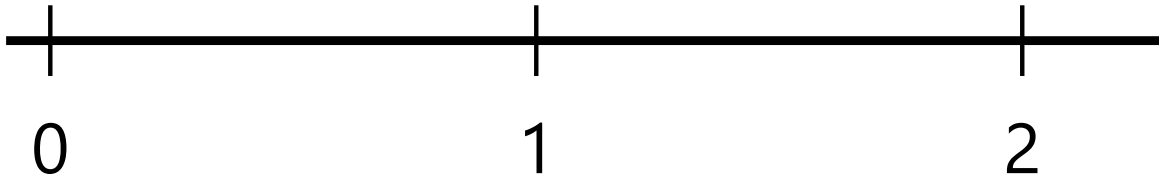
Concept: Operations that preserve the solution set

Concept difficulty: Medium concept

Predict

Adding 5 to both sides of an equation loses nothing. Predict whether the same holds for multiplying both sides by 5, and for multiplying both sides by x .

Domain: the rationals $\cdot x$ (dimension



Declared domain

Attempt

From $x \cdot x = x$, first list every rational solution by inspection. Then divide both sides by x and solve the result. Compare the two solution sets and name the value of x that was silently ruled out by the division.

- Start with $x \cdot x = x$ over the rationals.
- State any condition before each division step.

Boundary test

Square both sides of $x = 3$. What are the solutions now, and what does that tell you about squaring?

An exact solution is not its decimal approximation

Subject: Mathematics

Topic: Linear equations

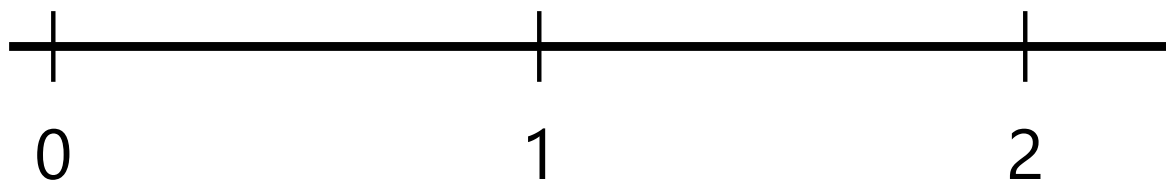
Concept: An exact solution is not its decimal approximation

Concept difficulty: Medium concept

Predict

A calculator gives 2.3333333 for x in $3x + 2 = 9$. Predict whether putting that number back in returns exactly 9, and if not, how far off it lands.

Domain: the rationals · x (dimension



Declared domain

Attempt

Solve $3x + 2 = 9$ over the rationals. Write the answer as a fraction in lowest terms. Then write the 4-decimal truncation of that fraction, substitute both into the equation, and state the difference each leaves on the left-hand side.

- Solve $3x + 2 = 9$ over rational x , then compare exact and truncated values.

Boundary test

Solve $4x = 2$ and write the answer as a decimal. Does the same objection apply?

Mathematics · Linear equations in one unknown